

Muhammad Rizwan

🏠 Lahore Punjab, Pakistan | 📞 +92-307-5250203 | ✉️ rixxwann@gmail.com |  |  | 

ABOUT

Motivated and high-achieving data science graduate with a strong foundation in research, machine learning, and programming. Gold Medalist with a proven track record of academic excellence and a passion for advancing knowledge in *medical image analysis, big data analytics, computer vision, large language models, federated learning, model fairness, and explainable AI*. Skilled in designing innovative research methodologies, solving complex problems, and contributing to impactful research in academia and interdisciplinary domains.

EDUCATION

Fast NUCES, Lahore Unlearn at Once (MS-Thesis)

MS Data Science | CGPA 3.87/4.00

Aug 2022 – June 2024

- * Developed two novel machine unlearning techniques—USI(F)R and DNF—that significantly improve the efficiency and reliability of data removal from machine learning models.
- * Advanced privacy-preserving and trustworthy AI by addressing key data management challenges and contributing new methodologies for building more secure and compliant AI systems.

Superior University COVID-19 fake news detection (FYP)

BS Computer Science | CGPA 3.48/4.00

Aug 2018 – Aug 2022

- * Developed an NLP-based system for COVID-19 fake news detection, incorporating preprocessing, TF-IDF, word embeddings, and transformer models (e.g., BERT) to extract meaningful textual features and train effective misinformation classification models.
- * Evaluated model performance using metrics such as accuracy and F1-score, demonstrating the system's potential to reduce the spread of misleading pandemic-related content on social media.

RESEARCH & PUBLICATIONS

1. **Muhammad Rizwan**; Hamza Naveed; Muhammad Zulqarnain, "*Comparative Analysis of Neural Network Architectures for Image Classification*", Journal of Intelligent Decision Making and Information Science 2 (2025): 456-471 <https://doi.org/10.59543/jidmis.v2i.11481>
2. **Muhammad Rizwan**; Maida Sajid; Salman Muneer; Muhammad Adnan Khan, "*Cross-Lingual Transfer Learning in Sign Language Recognition-Adapting American Sign Language Models to Pakistan Sign Language*", The Computer Journal, [Under Review]
3. Maida Sajid; **Muhammad Rizwan**; Salman Muneer; Muhammad Adnan Khan, "*Development of SHAP Weighted Website Performance Index for Website Benchmarking and Optimization*", Scientific Reports, [Submitted]
4. Sumaira Rafique; **Muhammad Rizwan**; Salman Muneer; Muhammad Adnan Khan, "*Improving Security of Federated Conditional Generative Adversarial Networks (GANs) through Loss-Based Identification of Label Flipping Attacks*", SN Computer Science, [Submitted]
5. Salman Muneer; Muhammad Fiaz; **Muhammad Rizwan**; Hamza Muneer; Ateeq Ur Rehman; Seada Hussen; Muhammad Adnan Khan, "*Integrating Advanced Machine learning and an Explainable Artificial Intelligence Approach with a Smart Energy Management System for Building Load Optimization*", International Journal of Energy Research, [Submitted]
6. **Muhammad Rizwan**, Mohsin Akram, Maida Sajid, Salman Muneer, Muhammad Adnan Khan, "*Hybrid Encryption-Based Secure Federated Learning for Healthcare: A Case Study on Brain Tumour MRI*", Health Information Science and Systems, [Submitted]
7. **Muhammad Rizwan**; Madiha Ijaz; Salman Muneer; Muhammad Adnan Khan, "*Role-Aware Sentiment Trajectory Analysis of Customer-Agent Interactions with Automated Highlights and Training Insights*", [Ready to Submit]
8. Maida Sajid; **Muhammad Rizwan**, "*IoT-Driven Predictive Modelling for Heart Disease Detection: A Survey of Remote Healthcare Monitoring*", [Ongoing]

TECHNICAL SKILLS

Languages : Python, SQL (Postgres, MS-SQL, MySQL), Java

Libraries : Pytorch, Keras, Transformers, Darts, Numpy, Matplotlib, Seaborn, Plotly, BuetifulSoup, spaCy, Gensim, NLTK, Langchain, OpenCV, Supervision, DeepFace

Data Tools : KNIME, Pentaho, Pyspark, dbt, PowerBI, Tableau

DevOps and API Tools : Git, GitLab, Airflow, Docker, Postman

Cloud : AWS, GCP, Snowflake

Others : Jira, Slack, Google Meet, Zoom, MS Office, GCR

PROJECTS

1. ATTENDANCE SYSTEM

- Develop an AI-based attendance system for efficient and automated tracking of personnel attendance
- The DeepFace library facilitates face recognition, with recognized faces triggering the upload of payloads to an S3 bucket. Subsequently, a Lambda function is triggered to store the data in DynamoDB.

2. DEMAND FORECASTING

- The garment store aims to accurately predict the monthly demand for its products to optimize inventory management
- Ten distinct models were trained using Darts for a univariate problem, and performance evaluation relied on RMSE metrics to assess predictive accuracy.

3. LOAD FORECASTING

- Built XGBoost, LSTM, and Transformer models for short-term load forecasting, applying robust hyperparameter tuning using Bayesian Optimization to optimize accuracy and generalization.
- Evaluated and compared model performance across multiple metrics, ensuring stable predictions under varying load patterns and seasonal fluctuations.

4. SILVER TREE BRANDS

- Use Google Big Query to fetch the data, do some transformation by taking guidance from KNIME Legacy workflows.
- Make the connection between Dataform and write the SQLX scripts for performing the transformation on the data, and save the results to the BQ datasets.

HONORS AND AWARDS

MEDAL DISTRIBUTION CEREMONY (Gold Medal) : National University of Emerging Sciences and Technology
SUPERIOR MERIT BASED SCHOLARSHIP : Superior University of Lahore
PBTE SCHOLASHIP :Punjab Board of Technical Education
KHADIM-E-PUNJAB UJAALA PROGRAM : Government of The Punjab

CERTIFICATIONS

Python	Hacker Rank	June 2022
SQL For Data Science	Great Learning	July 2022
Machine Learning	Coursera Stanford.AI	Feb 2023
Practical Data Science	Coursera Stanford.AI	April 2023
Data Analytics(Python)	Coursera IBM	June 2023

EXPERIENCE

Associate Lecturer	University Of Central Punjab	Oct 2024 – Present
- Designed course outlines, prepared and delivered lectures in Data Science, Machine Learning, and Analytics, and assessed student performance through assignments, projects, and examinations. - Fostered a positive and encouraging learning environment, helping students achieve significant academic improvements and confidence		
Associate Data Scientist	Reallytics.ai	July 2023 – July 2024
- Led comprehensive data exploration, revealing key patterns and trends to inform strategic decision-making - Developed and optimized machine learning models for impactful solutions within cross-functional teams. - Collaborates seamlessly with the web team to sweeten the user experience by integrating and enhancing AI products.		

REFERENCE

Dr. Asma Ahmad Farhan	Associate Professor asma.ahmad@nu.edu.pk	NUCES, Lahore
Dr. Muhammad Amjad Iqbal	Dean, Associate Professor amjad.iqbal@ucp.edu.pk	UCP, Lahore